PROJECT REPORT

TASK 27: DISTRIBUTION ANALYSIS USING PYTHON

1. Abstract

This project focuses on analyzing the distribution of numerical data using Python, Pandas, NumPy, and Matplotlib.

A sample student marks dataset containing 30 observations is used to generate a histogram and examine the concentration, spread, and skewness of student marks.

Descriptive statistical measures, including mean, median, standard deviation, minimum, maximum, and range, are calculated to support the distribution analysis.

2. Introduction

Distribution analysis is an important part of statistics and exploratory data analysis. It helps understand how numerical observations are spread across different value ranges.

A histogram is a graphical representation that groups numerical observations into intervals called bins. It displays the frequency of values in each interval.

By examining a histogram, analysts can identify where values are concentrated, how widely values are spread, and whether a distribution appears symmetric or skewed.

This project uses a student marks dataset to demonstrate these concepts.

3. Objectives

The main objectives are:

- To create a histogram for a numerical variable.
- To understand the frequency distribution of student marks.
- To experiment with different histogram bin sizes.
- To calculate descriptive statistical measures.
- To identify concentration and spread.
- To examine possible skewness.
- To record at least three observations from the data.

4. Tools and Technologies

Python: Used to implement the analysis.

Pandas: Used to create, load, and process the dataset.

NumPy: Used for numerical operations and histogram bin definitions.

Matplotlib: Used to generate the histogram.

Google Colab: Used as the Jupyter Notebook execution environment.

CSV: Used to store the dataset and statistical summaries.

5. Dataset Description

The project uses a manually created dataset containing 30 fictional student records.

The dataset consists of two columns:

Column	Description
student_id	Unique student identifier
marks	Marks obtained by the student

The student marks range from 35 to 96.

The dataset is created for educational and statistical demonstration.

6. Methodology

The project follows these steps:

1. Import the required Python libraries.
2. Create the student marks dataset.
3. Convert the dataset into a Pandas DataFrame.
4. Export and load the CSV dataset.
5. Inspect the dataset structure.
6. Calculate descriptive statistical measures.
7. Create a histogram using Matplotlib.
8. Add mean and median reference lines.
9. Experiment with different histogram bin sizes.
10. Calculate frequency in each marks interval.
11. Analyze concentration, spread, and skewness.
12. Record observations based on the results.
13. Export the histogram and statistical summaries.

7. Descriptive Statistical Analysis

The project calculates the following statistical measures:

Measurement	Value
Total students	30
Mean	67.50
Median	68.00
Standard deviation	13.25
Minimum marks	35
Maximum marks	96
Range	61
Skewness	-0.289

The mean represents the average marks, while the median represents the central value of the sorted observations.

The standard deviation measures how widely the observations are distributed around the mean.

The range represents the difference between the maximum and minimum values.

8. Histogram Generation

The histogram is generated using Matplotlib's `plt.hist()` function.

The marks are grouped into seven intervals of 10 marks each.

The horizontal axis represents student marks, and the vertical axis represents the number of students in each interval.

Vertical reference lines are included to show the mean and median values.

Insert the actual marks_distribution_histogram.png chart here.

9. Histogram Bin Analysis

The project compares histograms containing five and ten bins.

A smaller number of bins provides a broad overview of the data, whereas a larger number of bins provides more detailed information about individual intervals.

The main histogram uses seven bins with intervals of 10 marks, which provides an understandable representation of the sample dataset.

10. Frequency Distribution

The frequency distribution is:

Marks Interval	Number of Students
30 – 39	1
40 – 49	2
50 – 59	3
60 – 69	11
70 – 79	8
80 – 89	3
90 – 99	2

The highest concentration occurs in the 60–69 interval, followed by the 70–79 interval.

A total of 19 students, or 63.33% of the dataset, scored between 60 and 79.

11. Distribution Analysis

11.1 Concentration

Most students scored between 60 and 79.

This indicates that the majority of the observations are concentrated in the middle ranges of the dataset.

11.2 Spread

The marks range from 35 to 96, resulting in a range of 61.

The standard deviation is approximately 13.25, indicating the spread of marks around the mean.

11.3 Skewness

The calculated skewness is approximately -0.289.

A negative skewness value indicates that the distribution has a slightly longer tail towards lower values.

Since the value is relatively close to zero, the sample distribution does not show strong skewness.

11.4 Mean and Median Comparison

The mean is 67.50, while the median is 68.00.

Their closeness suggests that the distribution is approximately balanced around its centre, with a slight negative skew.

12. Key Observations

The project produces the following observations:

1) Nineteen out of 30 students, or 63.33%, scored between 60 and 79.
2) The mean marks are 67.50, while the median marks are 68.00, indicating that the central values are very close.
3) The skewness value of -0.289 indicates slight negative skewness.
4) The standard deviation of 13.25 indicates the spread of marks around the mean.
5) Marks range from 35 to 96, demonstrating variation in student performance.
6) The 60–69 interval has the highest frequency, containing 11 students.

13. Importance of Distribution Analysis

Distribution analysis helps analysts understand numerical data before applying more advanced statistical or machine-learning methods.

It helps identify the concentration of observations, detect possible skewness, understand variability, and recognize unusual values.

Histograms provide an intuitive way to communicate numerical distributions.

14. Applications

Distribution analysis is useful in:

- Student performance analysis.
- Examination result analysis.
- Salary distribution analysis.
- Sales and revenue analysis.
- Healthcare data analysis.
- Exploratory data analysis.
- Statistical and machine-learning workflows.

15. Limitations

The project uses a small manually created dataset containing 30 observations.

The results describe only the sample dataset and should not be interpreted as actual academic performance statistics.

The histogram provides a basic distribution analysis and does not establish relationships between different variables.

16. Future Enhancements

The project could be extended by:

- Using larger real-world datasets.
- Comparing distributions between different student groups.
- Identifying statistical outliers.
- Creating box plots and density plots.
- Comparing multiple numerical variables.
- Applying advanced statistical analysis.

17. Conclusion

This project demonstrates how Python, Pandas, NumPy, and Matplotlib can be used to analyze numerical data distributions.

A histogram is created to visualize student marks, and descriptive statistics are calculated to understand concentration, spread, and skewness.

The analysis shows that most student marks are concentrated between 60 and 79, with a slight negative skew.

This exercise provides practical knowledge of statistical analysis and data visualization.

18. References

1. Pandas Documentation: https://pandas.pydata.org/docs/
2. NumPy Documentation: https://numpy.org/doc/
3. Matplotlib Documentation: https://matplotlib.org/stable/
4. Matplotlib Histogram Reference: https://matplotlib.org/stable/api/_as_gen/matplotlib.pyplot.hist.html

19. Project Links

GitHub Repository: https://github.com/venky1318/Distribution_Analysis